

-- Source page 1 --

Discrete Mathematics**Ex 12.1****Question 1.**

Determine whether $*$ is a binary operation on the sets given below

(i) $a*b=a \cdot |b|$ on $\mathbb{R}$,

(ii) $a*b=\min(a,b)$ on $A=\{1,2,3,4,5\}$

(iii) $(a*b)=a\sqrt{b}$ is binary on $\mathbb{R}$.

Solution:

(i) Yes.

Reason: $a, b \in \mathbb{R}$

$\in$. So, $|b| \in \mathbb{R}$

$\in$ when $b \in \mathbb{R}$

$\in$

Now multiplication is binary on $\mathbb{R}$

So $a|b| \in \mathbb{R}$ when $a, b \in \mathbb{R}$.

(Le.) $a*b \in \mathbb{R}$.

$*$ is a binary operation on $\mathbb{R}$.

(ii) Yes.

Reason: $a, b \in \mathbb{R}$ and minimum of (a, b) is either a or b but $a, b \in \mathbb{R}$.

So, $\min(a, b) \in \mathbb{R}$.

(Le.) $a*b \in \mathbb{R}$

$\in$.

$*$ is a binary operation on $\mathbb{R}$.

(iii) $a*b=a\sqrt{b}$ where $a, b \in \mathbb{R}$

$\in$.

No. $*$ is not a binary operation on $\mathbb{R}$.

Reason: $a, b \in \mathbb{R}$

$\in$.

b

$\Rightarrow$ can be -ve number also and square root of a negative number is not real.

So $\sqrt{b}$

$\in \mathbb{R}$

$\notin$ even when $b \in \mathbb{R}$

$\in$.

So $\sqrt{b}$

$\in \mathbb{R}$

$\notin$. ie., $a*b \notin \mathbb{R}$

$\notin$.

$*$ is not a binary operation on $\mathbb{R}$.

Question 2.

On $\mathbb{Z}$, define $\otimes$ by $(m \otimes n) = mn + nm: \forall m, n \in \mathbb{Z}$. Is $\otimes$ binary on $\mathbb{Z}$?

Solution:

No. $*$ is not a binary operation on $\mathbb{Z}$.

Reason: Since $m, n \in \mathbb{Z}$.

So, m, n can be negative also.

Now, if n is negative (Le.) say $n =$

k

- where k is +ve.

Then $mn = m - k = 1mk \notin \mathbb{Z}$.

Similarly, when m is negative then $nm \notin \mathbb{Z}$

$\notin$. Therefore $m \otimes n \notin \mathbb{Z}$.

$*$ $\notin$

$\Rightarrow *$ is not a binary operation on $\mathbb{Z}$.

Question 3.

Let $*$ be defined on $\mathbb{R}$ by $(a * b) = a + b + ab$

1

– Is $*$ binary on $\mathbb{R}$? If so, find $3 * (-715)$

715

–

)

Solution:

$a * b = a + b + ab$

7

–

Now when $a, b \in \mathbb{R}$

$\in$, then $ab \in \mathbb{R}$

$\in$ also $a + b \in \mathbb{R}$

$\in$,

So, $a + b + ab \in \mathbb{R}$.

We know

$7 \in \mathbb{R}$

– $\in$.

So, $a + b + ab$

7

– $\in \mathbb{R}$.

-- Source page 2 --

(ie.) $a * b \in \mathbb{R}$.

So, $*$ is a binary operation on $\mathbb{R}$.

Now, to find $3 * (-715)$

$3 * (-$

715

–

) = 3 + (

715

–

) + (3)(

715

–

)

$7 = 3$

715

2115

$7 = 45$

7

21

10515 =

8815

–

–

–

–

--

–

–

Question 4.

Let $A = \{a + \sqrt{5}b : a, b \in \mathbb{Z}\}$

∈

. Check whether the usual multiplication is a binary operation on A.

Solution:

Let $A = \{a + \sqrt{5}b \mid a, b \in \mathbb{Z}\}$ and $B = \{c + \sqrt{5}d \mid c, d \in \mathbb{Z}\}$

∈

Now $A * B = \{a + \sqrt{5}b\} \{c + \sqrt{5}d\}$

$= \{ac + \sqrt{5}ad + \sqrt{5}bc + \sqrt{5}b\sqrt{5}d\}$

$= \{(ac + 5bd) + \sqrt{5}(ad + bc)\}$

∈

Where $a, b, c, d \in \mathbb{Z}$

∈

So $*$ is a binary operation.

Question 5.

(i) Define an operation $*$ on $\mathbb{Q}$ as follows: $a * b = (a + b)^2$; $a, b \in \mathbb{Q}$

∈

. Examine the closure,

commutative, and associative properties satisfied by $*$ on $\mathbb{Q}$.

(ii) Define an operation $*$ on $\mathbb{Q}$ as follows: $a * b = (a + b)^2$; $a, b \in \mathbb{Q}$

∈

. Examine the existence of

identity and the existence of inverse for the operation $*$ on $\mathbb{Q}$.

Solution:

(i) 1. Closure property:

Let $a, b \in \mathbb{Q}$.

Now $a * b = (a + b)^2 \in \mathbb{Q}$. (Since $a, b \in \mathbb{Q}$)

So, closure property is satisfied.

2. Commutative property:

Let $a, b \in \mathbb{Q}$

∈

Now $a * b = (a + b)^2$

$b * a = (b + a)^2$

$(1) = (2) \Rightarrow a * b = b * a$

$\Rightarrow$ Commutative property is satisfied.

3. Associative property:

Let $a, b, c \in \mathbb{Q}$

To prove associative property we have to prove that $a ($

$* b) c = (a$

$*$

$* b) c$

$*$

LHS: $a ($

$* b) c$

$*$

Now $b * c = (b + c)^2 = D$ (say)

So $a ($

$* b * c) = a D =$

$*$

$a + D^2 = a + (b + c)^2 = a + b^2 + 2bc + c^2$

RHS: $(a * b) * c$

$a * b = (a + b)^2 = K$ (say)

So $(a * b) * c = K * c = (K + c)^2 = a^2 + b^2 + 2ab + c^2 + 2ac + 2bc$

$(1) \equiv (2)$

So, associative property is not satisfied.

(ii) $a * b = a + b^2$

Let $e \in Q$ be the identity element. Then $a * e = a$

To find $e: a * e = a$

(i.e.) $a + e^2 = a \Rightarrow a + e = 2a \Rightarrow e = 2a - a$

-- Source page 3 --

(i.e.) the identity element $e = a$ which is not possible.

So, the identity element does not exist and so inverse does not exist.

Question 6.

Fill in the following table so that the binary operation $*$ on $A = \{a, b, c\}$ is commutative.

$*$	a	b	c
a	b		
b	c	b	a
c	a		c

Solution:

Given that the binary operation $*$ is Commutative. To find $a * b : a * b = b * a$

*

Since $*$ is a
Commutative)

Here $b * a = c$. So $a * b = c$ To find $a *$

$*$	a	b	c
a	b		
b	c	b	a
c	a		c

$a * c = c * a$

*

*

Since $*$ is a Commutative)

$c * a = a$. (Given)

So $a * c = a$

*

To find $c * b :$

$c * b = b * c$

*

Here $b * c = a$

*

.

So $c * b = a$

Question 7.

Consider the binary operation $*$ defined on the set $A = \{a, b, c, d\}$ by the following table:

*	a	b	c	d
a	a	c	b	d
b	d	a	b	c
c	c	d	a	a
d	d	b	a	c

- Is it commutative and associative?

Solution:

From the table

$$b * c = bc$$

*

$$* b = d$$

So, the binary operation is not commutative.

To check whether the given operation is associative.

Let $a, b, c \in A$

$\in$

To prove the associative property we have to prove that $a ($

$$* b * c) = (a$$

*

$$* b) * c$$

*

From the table,

$$\text{LHS: } b * c = b$$

-- Source page 4 --

So, $a ($

$$* b * c) = a * b = c$$

$$\text{RHS: } a * b = c$$

$$\text{So, } (a * b) * c = c * a = a$$

*

*

(1) $\boxtimes$ (2). So, $a ($

$$* b * c) \neq (a * b) * c$$

Therefore The binary operation is not associative.

Question 8.

Question 8.

Let $A = \begin{pmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & 0 & 1 \end{pmatrix}$, $B = \begin{pmatrix} 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \\ 1 & 0 & 0 & 1 \end{pmatrix}$, $C = \begin{pmatrix} 1 & 1 & 0 & 1 \\ 0 & 1 & 1 & 0 \\ 1 & 1 & 1 & 1 \end{pmatrix}$ be any three boolean matrices of the same type. Find (i) $A \vee B$ (ii) $A \wedge B$ (iii) $(A \vee B) \wedge C$ (iv)

$(A \wedge B) \vee C$. Solution:

$$(i) \quad A \vee B = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 1 & 0 & 0 & 1 \end{pmatrix}$$

$$(ii) \quad A \wedge B = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 1 \end{pmatrix}$$

$$A \vee B = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 1 & 0 & 0 & 1 \end{pmatrix}$$

$$(iii) \quad (A \vee B) \wedge C = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 1 & 0 & 0 & 1 \end{pmatrix} \wedge \begin{pmatrix} 1 & 1 & 0 & 1 \\ 0 & 1 & 1 & 0 \\ 1 & 1 & 1 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 1 & 0 & 1 \\ 0 & 1 & 1 & 0 \\ 1 & 0 & 0 & 1 \end{pmatrix}$$

(iv)

$$(iv) \quad A \wedge B = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 1 \end{pmatrix}$$

$$\text{So, } (A \wedge B) \vee C = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 1 \end{pmatrix} \vee \begin{pmatrix} 1 & 1 & 0 & 1 \\ 0 & 1 & 1 & 0 \\ 1 & 1 & 1 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 1 & 0 & 1 \\ 0 & 1 & 1 & 0 \\ 1 & 1 & 1 & 1 \end{pmatrix}$$

Question 9.

(i) Let $M = \{(xxx): x \in \mathbb{R}\}$
 $\{0\}$

and let $*$ be the matrix multiplication. Determine whether M is closed under $*$. If so, examine the commutative and associative properties satisfied by $*$ on M .

(ii) Let $M = \{(xxx): x \in \mathbb{R}\}$
 $\{0\}$

and let $*$ be the matrix multiplication. Determine whether M is closed under $*$. If so, examine the existence of identity, existence of inverse properties for the operation $*$ on M .

Solution:

(i) $M = \{(xxx): x \in \mathbb{R}\}$
 $\{0\}$

$\in$

$=$

*** Matrix multiplication**

Let $A = (xxx), B = (yyy) \in M$

$\in$

Where $x, y \in \mathbb{R}$

$\{0\}$

$\in$

Now $AB = (xxx)(yyy) = (2xy \ 2xy \ 2xy) \in M$

$\in$

Since $2xy \in \mathbb{R}$

$\in M$

$\in$

So, M is closed under $*$.

To verify the existence of identity property:

; Let $E = (eee) \in M$

$\in$

such that

$A \cdot E = A$

$*$

$* A = A$

Where $A = (xxx)$

To find $E \cdot A = A$

$*$

$\Rightarrow (xxx)(eee) = (xxx)$ i.e., $(2xe \ 2xe \ 2xe) = (xxx)$

2

$\Rightarrow xe = x \Rightarrow e = x/2x = 1/2$ Therefore $E = (1/2 \ 1/2 \ 1/2) \in M$

$\in M$

$\in$

So, identity property is satisfied.

To verify the inverse property:

-- Source page 5 --

Let A

1

$\in M$

$\in$

Such that $A \cdot A$

1

$\in M$

1

 $\rightarrow A=E$

To find A

1

- :

 $A * A$

1

 $\rightarrow E$ i.e., (xxxx) $(yyyy) = (1/21/21/21/2) \Rightarrow (2xy2xy2xy2xy) = (1/21/21/21/2)$

2

 $\Rightarrow xy = 1/2$

2

 $\Rightarrow y = 12(2x) = 14x$

Therefore A

1

 $\rightarrow (1/4 \times 1/4 \times 1/4 \times 1/4 \times)$

M

 $\in$

So, inverse property is satisfied.

Question 10.(i) Let A be $Q \setminus \{1\}$

☒

. Define * on A by $x*y = x+y-xy$. Is * binary on A? If so, examine the commutative and associative properties satisfied by * on A.(ii) Let A be $Q \setminus \{1\}$

☒

. Define * on A by $x*y = x+y-xy$. Is * binary on A? If so, examine the existence of identity, existence of inverse properties for the operation * on A.**Solution:**(i) Let $a, b \in A$ (i.e.) $a \neq 1, b \neq 1$ Now $a*b = a+b-ab$ If $a+b$ $ab=1$ $a+b$ ab $1=0$

-

 $\Rightarrow$

-

-

(i.e.) $a(1-b)$ $1(1$

-

 $-b)=0$

(a

 $1)(1$

-

 $-b)=0 \Rightarrow a=1, b=1$ But $a \neq 1, b \neq 1$

So (a

 $1)(1$ $6) \neq 1$

-

-

(i.e.) $a * b \in A$

∈

. So $*$ is a binary on A .

To verify the commutative property:

Let $a, b \in A$ (i.e.) $a \neq 1, b \neq 1$

Now $a * b = a + b - ab$

and $b * a = b + a - ba$

ba

*

-

So $a * b = b * a$

$*$ is commutative on A .

To verify the associative property:

Let $a, b, c \in A$

∈

(i.e.) $a, b, c \neq 1$

To prove the associative property we have to prove that

$a * (b * c) = (a * b) * c$

$a * (b * c) = (a * b) * c$

*

$a * b) * c$

*

LHS: $b * c = b + c - bc$

$bc = D$

*

-

(say)

So $a * (b * c) = a * (b + c - bc)$

$a * (b * c) = a * (b + c - bc)$

$= a + (b + c - bc) - a(b + c - bc)$

$= a + b + c - bc - ab - ac + abc = a + b + c - ab - bc - ac + abc$ RHS: $(a * b) * c = (a + b - ab) * c$ (say) S

$(a * b) * c = (a + b - ab) * c = (a + b - ab) + c - (a + b - ab)c$

$(a + b - ab)c = a + b - ab + c - ac - bc + abc = a + b + c - ab - bc - ac + abc$

(ii) To verify the identity property:

Let $a \in A$ ($a \neq 1$)

If possible let $e \in A$

∈

such that

$a * e = a$ and $e * a = a$

*

*

To find e :

$a * e = a$

*

(i.e.) $a + e - ae = a$

$ae = e$

-

$\Rightarrow e(1 - a) = 0 \Rightarrow e = 0$ (Since $a \neq 1$)

So, $e = 0 \in A$

∈

(i.e.) Identity property is verified.

To verify the inverse property:

-- Source page 6 --

Let $a \in A$ (i.e. $a \neq 1$)

If possible let $a' \in A$

∈

such that

To find a' :

$$a * a' = e \text{ (i.e.) } a + a'$$

$$aa$$

-

$$\emptyset = 0$$

a

$$\Rightarrow (1$$

$$a) =$$

a

-

$$\Rightarrow a' = -a1 - a = aa$$

$$1 A($$

$$- \in$$

Since $a \in 1$ So, $a \in A$

$$\in$$

$\Rightarrow$ For every $a \in A$

$$\in$$

there is an inverse $a' \in A$

$$\in$$

such that $a * a' = a' * a = e$

$\Rightarrow$ Inverse property is verified.

Ex 12.2

Question 1.

Let p : Jupiter is a planet and q : India is an island be any two simple statements. Give verbal sentence describing each of the following statements.

$$(i) \neg p$$

$$(ii) p \neg$$

$$\wedge q$$

$$(iii) \neg p \vee q$$

$$(iv) p$$

$\neg$

$$\rightarrow q$$

$$(v) p \leftrightarrow q$$

Solution:

(i) $\neg p$: Jupiter is not a planet

$$(ii) p \neg$$

$\wedge q$: Jupiter is not a planet and India is not an island

(iii) $\neg p \vee q$: Jupiter is not a planet or India is an island.

$$(iv) p$$

$\neg$

$\rightarrow q$: If Jupiter is a planet then India is not an island

(v) $p \leftrightarrow q$: If Jupiter is a planet if and only if India is an island

Question 2.

Write each of the following sentences in symbolic form using statement variables p and q .

(i) 19 is not a prime number and all the angles of a triangle are equal.

(ii) 19 is a prime number or all the angles of a triangle are not equal

(iii) 19 is a prime number and all the angles of a triangle are equal

(iv) 19 is not a prime number

Solution:

p : 19 is a prime number

q : All the angles of a triangle are equal

$$(i) \neg p \wedge q$$

$$(ii) p \neg$$

$\vee q$

(iii) $p \wedge q$

(iv) $\neg p$

Question 3.

Determine the truth value of each of the following statements

(i) If $6+2=5$, then the milk is white.

(ii) China is in Europe or $\sqrt{3}$ is an integer

(iii) It is not true that $5+5=9$ or Earth is a planet

(iv) 11 is a prime number and all the sides of a rectangle are equal

Solution:

(i) $p, \neg q$ is True : T

(ii) $p \vee q$ is False: F

(iii) $\neg p \vee q$ is True : T

-- Source page-7 --

(iv) $p \wedge q$ is False: F

Question 4.

Which one of the following sentences is a proposition?

(i) $4+7=12$

(ii) What are you doing?

(iii) $3n$

$81, n \in \mathbb{N}$

$\leq$

$\in$

(iv) Peacock is our national bird

(v) How tall this mountain is!

Solution:

(i) is a proposition

(ii) not a proposition

(iii) is a proposition

(iv) is a proposition

(v) not a proposition

Question 5.

Write the converse, inverse, and contrapositive of each of the following implication.

(i) If x and y are numbers such that $x=y$, then $x^2=y^2$

(ii) If a quadrilateral is a square then it is a rectangle

Solution:

(i) Converse: If x and y are numbers such that $x^2=y^2$ then $x=y$.

Inverse: If x and y are numbers such that $x \neq y$ then $x^2 \neq y^2$.

Contrapositive : If x and y are numbers such that $x^2 \neq y^2$ then $x \neq y$.

(ii) Converse: If a quadrilateral is a rectangle then it is a square.

Inverse: If a quadrilateral is not a square then it is not a rectangle.

Contrapositive : If a quadrilateral is not a rectangle then it is not a square.

Question 6.

Construct the truth table for the following statements.

(i) $\neg p \neg$

$\wedge q$

(ii) $\neg(p \neg$

$\wedge q)$

(iii) $(p \vee q) \neg$

$\vee q$

(iv) $(\neg p \rightarrow r) ($

$\wedge p \leftrightarrow q)$

Solution:

(i) Truth table for $\neg p \neg$ $\wedge q$

p	q	$\neg p$	$\neg q$	$\neg p \wedge \neg q$
T	T	F	F	F
T	F	F	T	F
F	T	T	F	F
F	F	T	T	T

(ii) Truth table for $\neg(p \neg$ $\wedge q)$

p	q	$\neg q$	$p \wedge \neg q$	$\neg(p \wedge \neg q)$
T	T	F	F	T
T	F	T	T	F
F	T	F	F	T
F	F	T	F	T

(iii) Truth table for $(p \vee q) \neg$ $\vee q$

-- Source page 8 --

p	q	$p \vee q$	$\neg q$	$(p \vee q) \vee \neg q$
T	T	T	F	T
T	F	T	T	T
F	T	T	F	T
F	F	F	T	T

(iv) Truth table for $(\neg p \rightarrow r)$ $\wedge (p \leftrightarrow q)$

p	q	r	$\neg p$	$\neg p \rightarrow r$	$p \leftrightarrow q$	$(\neg p \rightarrow r) \wedge (p \leftrightarrow q)$
T	T	T	F	T	T	T
T	T	F	F	T	T	T
T	F	T	F	T	F	F
T	F	F	F	T	F	F
F	T	T	T	T	F	F
F	T	F	T	F	F	F
F	F	T	T	T	T	T
F	F	F	T	F	T	F

Question 7.

Verify whether the following compound propositions are tautologies or contradictions or contingency

(i) $(p \wedge q) \neg$ $\wedge$ $p \vee q)$

(ii) $((p \vee q) \neg$ $\wedge p) \rightarrow q$ (iii) $(p \rightarrow q)$ $(\neg$ $\leftrightarrow$ $p \rightarrow q)$ (iv) $((p \rightarrow q) ($ $\wedge q \rightarrow r))$ $($ $\rightarrow p \rightarrow r)$

Solution:

(i) Truth table for $(p \wedge q) \neg$ $\wedge$ $p \vee q)$

p	q	$p \wedge q$	$p \vee q$	$\neg(p \vee q)$	$(p \wedge q) \wedge \neg(p \vee q)$
T	T	T	T	F	F
T	F	F	T	F	F
F	T	F	T	F	F
F	F	F	F	T	F

In the above Truth table the last column entries are 'F'. So the given propositions is a contradiction.

(ii) Truth table for $((p \vee q) \neg$ $\wedge p) \rightarrow q$

p	q	$p \vee q$	$\neg p$	$(p \vee q) \wedge \neg p$	$((p \vee q) \wedge \neg p) \rightarrow q$
T	T	T	F	F	T
T	F	T	F	F	T
F	T	T	T	T	T
F	F	F	T	F	T

-- Source page 9 --

In the above truth table the last column entries are 'T'. So the given propositions is a tautology.

(iii) Truth table for $(p \rightarrow q)$ $(\neg$ $\leftrightarrow$ p $\neg$ $\rightarrow q)$

p	q	$\neg p$	$\neg q$	$p \rightarrow q$	$\neg p \rightarrow \neg q$	$(p \rightarrow q) \leftrightarrow (\neg p \rightarrow \neg q)$
T	T	F	F	T	T	T
T	F	F	T	F	T	F
F	T	T	F	T	F	F
F	F	T	T	T	T	T

In the above truth table the entries in the last column are a combination of 'T' and 'F'. So the given statement is neither propositions is neither tautology nor a contradiction. It is a contingency.

(iv) Truth table for $((p \rightarrow q) ($ $\wedge q \rightarrow r))$ $($ $\rightarrow p \rightarrow r)$

p	q	r	$p \rightarrow q$	$q \rightarrow r$	$(p \rightarrow q) \wedge (q \rightarrow r)$	$p \rightarrow r$	$((p \rightarrow q) \wedge (q \rightarrow r)) \rightarrow (p \rightarrow r)$
T	T	T	T	T	T	T	T
T	T	F	T	F	F	F	T
T	F	T	F	T	F	T	T
T	F	F	F	T	F	F	T
F	T	T	T	T	T	T	T
F	T	F	T	F	F	T	T
F	F	T	T	T	T	T	T
F	F	F	T	T	T	T	T

The last column entries are 'T'. So the given proposition is a tautology.

Question 8.

Show that (i) $\neg(p \wedge q)$

$\neg$

$\equiv p \neg$

$\vee q$

(ii) $\neg(p \rightarrow q) \equiv p \neg$

$\wedge q$

Solution:

(i) Truth table for $\neg(p \wedge q)$

p	q	$p \wedge q$	$\neg(p \wedge q)$
T	T	T	F
T	F	F	T
F	T	F	T
F	F	F	T

Truth table for $\neg p \neg$

$\vee q$

p	q	$\neg p$	$\neg q$	$\neg p \vee \neg q$
T	T	F	F	F
T	F	F	T	T
F	T	T	F	T
F	F	T	T	T

-- Source page 10 --

The entries in the column $\neg(p \wedge q)$ and $\neg p \neg$

$\vee q$ are identical. So, $\neg(p \wedge q)$

$\neg$

$\equiv p \neg$

$\vee q$

(ii) To prove $\neg(p \rightarrow q) \equiv p \neg$

$\wedge q$.

Truth table for $\neg(p \rightarrow q)$

p	q	$p \rightarrow q$	$\neg(p \rightarrow q)$
T	T	T	F
T	F	F	T
F	T	T	F
F	F	T	F

Truth table for $p \rightarrow$

$\wedge q$

p	q	$\neg q$	$p \wedge \neg q$
T	T	F	F
T	F	T	T
F	T	F	F
F	F	T	F

The entries in the column $\neg(p \rightarrow q)$ and $p \wedge \neg q$ are identical and hence they are equivalent.

Question 9.

Prove that $q \rightarrow p$

$\neg$

$\equiv p$

$\neg$

$\rightarrow q$

Solution:

Truth table for $q \rightarrow p$

q	p	$q \rightarrow p$
T	T	T
F	T	T
T	F	F
F	F	T

Truth table for $\neg p$

$\neg$

$\rightarrow q$

p	q	$\neg p$	$\neg q$	$\neg p \rightarrow \neg q$
T	T	F	F	T
T	F	F	T	T
F	T	T	F	F
F	F	T	T	T

The entries in the column corresponding to q

p

$\rightarrow$ and $\neg p$

$\neg$

$\rightarrow q$ are identical and hence they are equivalent.

Question 10.

Show that p

q

$\rightarrow$ and q

p

$\rightarrow$ are not equivalent

Solution:

Truth table for $p \rightarrow q$

-- Source page 11 --

p	q	$p \rightarrow q$
T	T	T
T	F	F
F	T	T
F	F	T

Truth table for $q \rightarrow p$

q	p	$q \rightarrow p$
T	T	T
F	T	T
T	F	F
F	F	T

The entries in the column corresponding to p

q

$\rightarrow$ and q

p

$\rightarrow$ are not identical, hence they are not equivalent.

Question 11.

Show that $\neg(p \leftrightarrow q) \equiv p$

$\neg$

$\leftrightarrow q$

Solution:

Truth table for $\neg(p \leftrightarrow q)$

p	q	$p \leftrightarrow q$	$\neg(p \leftrightarrow q)$
T	T	T	F
T	F	F	T
F	T	F	T
F	F	T	F

Truth table for p

$\neg$
 $\leftrightarrow q$

p	q	$\neg q$	$p \leftrightarrow \neg q$
T	T	F	F
T	F	T	T
F	T	F	T
F	F	T	F

Question 12.

Check whether the statement p

$(q$

$p)$

$\rightarrow$

$\rightarrow$

is a tautology or a contradiction without using the truth table. Solution:

p

$($

$\rightarrow q \rightarrow p)$

$\neg$

$\equiv p ($

$\vee q \rightarrow p)$ (Conditional equivalence)

$($

$\equiv q \rightarrow p) \neg$

$\vee p$ (Commutative

law)

$(\neg$

$\equiv$

$q \vee p) \neg$

$\vee p$ (Conditional equivalence)

$\neg$

$\equiv q ($

$\vee p \neg$

$\vee p)$ (Associative

law)

$\neg$

$\equiv q T$

$\vee$ (Complement law)

T

$\equiv$ (Identity law)

Therefore p

(
 $\rightarrow q \rightarrow p$) is a tautology.

Question 13.

Using truth table check whether the statements $\neg(p \vee q)$ ($\neg$
 $\vee$
 $p \wedge q$) and $\neg p$ are logically

-- Source page 12 --

equivalent. Solution:

Truth table for $\neg(p \vee q)$ ($\neg$

$\vee$

$p \wedge q$)

C ₁	C ₂	C ₃	C ₄	C ₅	C ₆	C ₇
p	q	$p \vee q$	$\neg(p \vee q)$	$\neg p$	$\neg p \wedge q$	$\neg(p \vee q) \vee (\neg p \wedge q)$ (C ₄) $\vee$ (C ₆)
T	T	T	F	F	F	F
T	F	T	F	F	F	F
F	T	T	F	T	T	T
F	F	F	T	T	F	T

In the above table C₇ and C₅ are identical

$\neg$ (

$\Rightarrow$

$p \vee q$) ($\neg$

$\vee$

$p \wedge q$)

$\neg$

$\equiv p$ is verified.

Question 14.

Prove p

(

$\rightarrow q \rightarrow r$)

(

$\equiv p \wedge q \rightarrow r$ without using truth table

Solution:

p

(

$\rightarrow q \rightarrow r$)

$\neg$

$\equiv p$ (

$\vee q \rightarrow r$) (Conditional equivalence)

(

$\equiv q \rightarrow r$) ($\neg$

$\vee$

p) (Commutative

law)

($\neg$

$\equiv$

$q \vee r$) ($\neg$

$\vee$

p) (Conditional equivalence)

($\neg$

$\equiv$

p) ($\neg$

$\vee$
 $q \vee r$ (Commutative law)

 $(\neg$
 $\equiv$
 $p \neg$
 $\vee q) \vee r$ (Associative law)

 $\neg($
 $\equiv$
 $p \wedge q) \vee r$ (Demorgan's law)

 $($
 $\equiv p \wedge q) \rightarrow r$ (Conditional equivalence) Therefore p

 $($
 $\rightarrow q \rightarrow r)$
 $($
 $\equiv p \wedge q) \rightarrow r$
Question 15.
Prove that p
 $(\neg$
 $\rightarrow$
 $q \vee r)$
 $\neg$
 $\equiv p (\neg$
 $\vee$
 $q \vee r)$ using truth table.

Solution:
Truth table for p
 $(\neg$
 $\rightarrow$
 $q \vee r)$

p	q	r	$\neg q$	$\neg q \vee r$	$p \rightarrow (\neg q \vee r)$
T	T	T	F	T	T
T	T	F	F	F	F
T	F	T	T	T	T
T	F	F	T	T	T
F	T	T	F	T	T
F	T	F	F	F	T
F	F	T	T	T	T
F	F	F	T	T	T

 Truth table for $\neg p (\neg$
 $\vee$
 $q \vee r)$

$\neg p$	q	r	$\neg p$	$\neg q$	$\neg q \vee r$	$\neg p \vee (\neg q \vee r)$
T	T	T	F	F	T	T
T	T	F	F	F	F	F
T	F	T	F	T	T	T
T	F	F	F	T	T	T
F	T	T	T	F	T	T
F	T	F	T	F	F	T
F	F	T	T	T	T	T
F	F	F	T	T	T	T

-- Source page 13 --

The entries in the column corresponding to p

$(\neg$

$\rightarrow$

$q \vee r)$ and $\neg p (\neg$

$\vee$

$q \vee r)$ are identical. Hence

they are equivalent.

Ex 12.3

Choose the correct or the most suitable answer from the given four alternatives.

Question 1.

A binary operation on a set S is a function from

(a) S

S

$\rightarrow$

(b) $(S \times S)$

S

$\rightarrow$

(c) S

$(S \times S)$

$\rightarrow$

(d) $(S \times S)$

$(S \times S)$

$\rightarrow$

Solution:

(b) $(S \times S)$

S

$\rightarrow$

Question 2.

Subtraction is not a binary operation in

(a) $\mathbb{R}$

(b) $\mathbb{Z}$

(c) $\mathbb{N}$

(d) $\mathbb{Q}$

Solution:

(c) $\mathbb{N}$

Hint:

For example $2, 5 \in \mathbb{N}$ but 2

$5 - 3$

$-$

$\notin \mathbb{N}$

Question 3.

Which one of the following is a binary operation on N ?

- (a) Subtraction
- (b) Multiplication
- (c) Division
- (d) All the above

Solution:

- (b) Multiplication

Question 4.

In the set R of real numbers ' $\otimes$ ' is defined as follows. Which one of the following is not a binary operation on R ?

- (a) $a \otimes b = \min(a, b)$
- (b) $a \otimes b = \max(a, b)$
- (c) $a \otimes b = a$
- (d) $a \otimes b = ab$

Solution:

- (d) $a \otimes b = ab$

Hint:

Since

$2, 1/2$

-

$\in R$, but (

2)

-

$1/2 \notin R$

Question 5.

The operation $*$ defined by $a * b = ab^7$ is not a binary operation on

- (a) Q^+
- (b) Z

-- Source page 14 --

(c) R

(d) C

Solution:

- (b) Z

Hint:

Since $3, 5 \in Z$, but $3 \times 5^7 \notin Z$.

Question 6.

In the set Q define $a \odot b = a + b + ab$. For what value of y , 3

(

$\odot y$

$5) = 7$

$\odot$

?

(a) $y = 23$

(b) $y =$

23

-

(c) $y =$

32

-

(d) $y = 4$

Hint: $a \otimes b = a + b + ab$

So, y

$5 =$

$\odot$

$y + 5 + 5y$

Now, 3

(

$\odot y$

$5) = 3$

$\odot$

$\odot x = 3 + x + 3x$

$= 3 + (5 + 6y) + 3(5 + 6y)$

$= 3 + 5 + 6y + 15 + 18y = 7$ (Given)

24

$\Rightarrow$

$y = 7$

23 =

16

-

-

$\Rightarrow y =$

16 24

-

= 23

-

Solution:

(b) $y =$

23

-

Question 7.

If $a * b = \sqrt{a^2 + b^2}$ on the real numbers then $*$ is

(a) commutative but not associative

(b) associative but not commutative

(c) both commutative and associative

(d) neither commutative nor associative

Let, $a, b \in \mathbb{R}$

$\in$

Now $a * b = \sqrt{a^2 + b^2} \Rightarrow b * a = \sqrt{b^2 + a^2} \Rightarrow a * b = b * a \Rightarrow *$ is commutative

Now to find a (

$*b * c)$

Now $b * c = \sqrt{b^2 + c^2} = D$ (Say)

So $a ($

$*b * c) = a * D =$

$*$

$\sqrt{a^2 + D^2}$

$= \sqrt{a^2 + b^2 + c^2}$

To find $(a * b) * c$

$(a * b) = \sqrt{a^2 + b^2} = E$ (Say)

Now $(a * b) * c = E * c = \sqrt{E^2 + c^2}$

$= \sqrt{a^2 + b^2 + c^2}$

$(1) = (2) \Rightarrow *$ is associative

So $*$ is both commutative and associative

Solution:

(c) both commutative and associative

Question 8.

Which one of the following statements has the truth value T ?

-- Source page 15 --

- (a) $\sin x$ is an even function.
- (b) Every square matrix is non-singular
- (c) The product of complex number and its conjugate is purely imaginary
- (d) $\sqrt{5}$ is an irrational number

Solution:

- (d) $\sqrt{5}$ is an irrational number

Question 9.

Which one of the following statements has truth value F ?

- (a) Chennai is in India or $\sqrt{2}$ is an integer
- (b) Chennai is in India or $\sqrt{2}$ is an irrational number
- (c) Chennai is in China or $\sqrt{2}$ is an integer
- (d) Chennai is in China or $\sqrt{2}$ is an irrational number

Solution:

- (c) Chennai is in China or $\sqrt{2}$ is an integer

Question 10.

If a compound statement involves 3 simple statements, then the number of rows in the truth table is

- (a) 9
- (b) 8
- (c) 6
- (d) 3

Solution:

- (b) 8

Hint:

(i.e.) $2^3=8$

Question 11.

Which one is the inverse of the statement $(p \vee q)$

(

$\neg p \wedge q$) ?

- (a) $(p \wedge q)$

(

$\neg p \vee q$)

- (b) $\neg(p \vee q)$

(

$\neg p \wedge q$)

- (c) $(\neg p \neg$

$\vee q)$

$(\neg$

$\rightarrow$

$p \neg$

$\wedge q)$

- (d) $(\neg p \neg$

$\wedge q)$

$(\neg$

$\rightarrow$

$p \neg$

$\vee q)$

Solution:

- (a) $(\neg p \neg$

$\wedge q)$

$(\neg$ $\rightarrow$ $p \neg$ $\vee q)$

Question 12 .

Which one is the contrapositive of the statement $(p \vee q) \rightarrow r$?(a) $\neg r$ $(\neg$ $\rightarrow$ $p \neg$ $\wedge q)$ (b) $\neg r$ $(\neg$ $\rightarrow$ $p \vee q)$ (c) r $($ $\rightarrow p \wedge q)$ (d) p $($ $\rightarrow q \vee r)$

Solution:

(a) $\neg r$ $(\neg$ $\rightarrow$ $p \neg$ $\wedge q)$

Question 13.

The truth table for $(p \wedge q) \neg$ $\vee q$ is given below

-- Source page 16 --

p	q	$(p \wedge q) \vee (\neg q)$
T	T	(a)
T	F	(b)
F	T	(c)
F	F	(d)

Which one of the following is true?

(a) (b) (c) (d)

(1) TTTT

(2) TFTT

(3) TTFT

(4) T,FFF

Hint: The truth table for $(p \boxtimes q) \neg$ $\vee q$

p	q	$p \wedge q$	$\neg q$	$(p \wedge q) \vee \neg q$
T	T	T	F	T
T	F	F	T	T
F	T	F	F	F
F	F	F	T	T

Solution:

(3) T T F T

Question 14.

In the last column of the truth table for $\neg(p \neg \vee q)$ the number of final outcomes of the truth value 'F' are (a) 1

(b) 2

(c) 3

(d) 4

Hint:

The truth table for $\neg(p \neg \vee q)$

p	q	$\neg q$	$p \vee \neg q$	$\neg(p \vee \neg q)$
T	T	F	T	F
T	F	T	T	F
F	T	F	F	T
F	F	T	T	F

Solution:

(c) 3

Question 15.

Which one of the following is incorrect? For any two propositions p and q , we have

(a) $\neg(p \vee q)$

$\neg$

$\equiv p \neg$

$\wedge q$

-- Source page 17 --

(b) $\neg(p \wedge q)$

$\neg$

$\equiv p \neg$

$\vee q$

(c) $\neg(p \vee q)$

$\neg$

$\equiv p \neg$

$\vee q$

(d) $\neg(\neg p) \equiv p$

Solution:

(c) $\neg(p \vee q)$ $\neg$ $\equiv p \neg$ $\vee q$

Question 16.

p	q	$(p \wedge q) \rightarrow \neg p$
T	T	(a)
T	F	(b)
F	T	(c)
F	F	(d)

Which of the following is correct for the truth $(p \wedge q)$ $\neg$ $\rightarrow p$?

, (a) (b) (c) (d)

(1) TTTT

(2) FTTT

(3) FTTT

(4) TTTF

Hint:

The truth table for $(p \wedge q)$ $\neg$ $\rightarrow p$

p	q	$p \wedge q$	$\neg p$	$(p \wedge q) \rightarrow \neg p$
T	T	T	F	F
T	F	F	F	T
F	T	F	T	T
F	F	F	T	T

Solution:

(2) FTTT

Question 17.

The dual of $\neg(p \vee q)$ [$\vee p$ ($\vee p \neg$ $\wedge r)$ is(a) $\neg(p \wedge q)$ [$\wedge p$ ($\vee p \neg$ $\wedge r)$ (b) $(p \wedge q)$ [$\wedge p$ ($\wedge p \neg$

$\vee r]$
 $(c) \neg(p \wedge q) [$
 $\wedge p ($
 $\wedge p \wedge r)]$
 $(d) \neg(p \wedge q) [$
 $\wedge p ($
 $\wedge p \neg$
 $\vee r)]$

Solution:

$(d) \neg(p \wedge q) \wedge p ($
 $\wedge p \neg$
 $\vee r)]$

Question 18.

The proposition $p (\neg$
 $\wedge$

$p \vee q)]$ is

- (a) a tautology
- (b) a contradiction
- (c) logically equivalent to $p \wedge q$

-- Source page 18 --

- (d) logically equivalent to $p \vee q$

Solution:

- (c) logically equivalent to $p \wedge q$

Question 19.

Determine the truth value of each of the following statements:

- (a) $4+2=5$ and $6+3=9$
- (b) $3+2=5$ and $6+1=7$
- (c) $4+5=9$ and $1+2=4$
- (d) $3+2=5$ and $4+7=11$

	(a)	(b)	(c)	(d)
(1)	F	T	F	T
(2)	T	F	T	F
(3)	T	T	F	F
(4)	F	F	T	T

Solution:

(1) F T F T

Question 20 .

Which one of the following is not true?

- (a) Negation of a negation of a statement is the statement itself.
- (b) If the last column of the truth table contains only T then it is a tautology.
- (c) If the last column of its truth table contains only F then it is a contradiction
- (d) If p and q are any two statements then p

q

$\leftrightarrow$ is a tautology.

Solution:

- (d) If p and q are any two statements then p

q

$\leftrightarrow$ is a tautology.